

the wider colour space setting of the digital camera will provide access to formerly unavailable colour data, resulting in better colours. The camera saves this 'camera-specific' information (like shutter speed, aperture, etc) in each image data file. A PIM compatible printer uses this information automatically while printing to ensure that it accurately reproduces the image captured by the camera and it does so by varying brightness, gamma and contrast settings accordingly.

The how of it

A conventional CCD sensor used in a digital camera consists of a multitude of tiny light-sensitive elements that convert light into an electrical signal. This in turn is converted to a dual or digital signal by an analog/digital converter module. These sensor elements do not recognise colour. So colour palettes of red, green and blue filters are attached on the CCD in such a manner that each individual light-sensitive element sees red, green or blue.

From the colour spectrum, the camera's electronics then calculate the 'correct' colours, a procedure referred to as colour interpolation. There are different colour spaces for different image processing hardware such as digital cameras. Therefore, 'correct' colours refer to that range of colours that can be used by the camera while other colours that cannot be reproduced by the camera are left out. The number of colours a camera can actually record is expressed by its colour depth value. Most digital cameras work with 24-bit colour, which means you can get 16 million colours. Some digital cameras can even process 30 bits,

which equals 1 billion colours.

To 'write' these colours, a specification called the YCbCr (Yellow, chrominance blue, chrominance red) colour space is used in digital cameras. This is a colour model by which colours are defined in terms of their brightness (referred to as luminance) and two of their shades (chrominance blue, chrominance red). Although the YCbCr colour space is usually used for storage in JPEG format, there are a few cameras that support TIFF YCbCr colour format.

Since the YCbCr colour space only permits 8 bits per component (Y, Cb and Cr), only 24 bits are stored with these file formats (i.e. 16 million colours). However, even 16 million colours is a large number—it's greater than the number of colours the human eye can recognise. Besides, you have special print colours such as light cyan and light magenta in addition to standard print colours such as cyan, magenta, yellow and black. Modern inkjet photo printers can apply most of these colours onto paper. But you can't always guarantee that the print perfectly replicates the real scene—what the digital camera 'sees' cannot be interpreted by the printer and you end up getting an image that might be close to the real thing but not exactly it.

So why is there a mismatch in the colours of the image captured by a digital camera and the colour print? The reason for this is that the colour management and colour optimisation technology available with conventional printers helps in optimising the colours only in the final image and cannot optimise the colours during the process of acquiring the image.

Peek-a-boo

Epson Stylus Photo

785EPX: This particular model is the first printer from Epson to include the PIM feature. Other features include compatibility with all popular digital camera memory card types such as CompactFlash Type I and II, IBM Microdrive, Sony Memory Stick, and Smart Media. It also features the patented BorderFree printing from Epson. This printer has an optional 1.6-inch colour preview monitor that attaches to the printer, letting you preview and select the images to be printed. The printer boasts of speeds of under a minute for a print size of 4x6 inches and lesser than 2 minutes for an 8x10 inch print size.

Epson CP-920Z: This 108x89x65-mm PIM-enabled digital camera weighs at 358 g, excluding the battery. It provides for image sizes of 2544x1940, 2048x1536, 1600x1200, 640x480 and supports image modes such as TIFF and JPEG. The camera has an optical viewfinder and auto white balance settings. It comes with a Compact Flash memory card with 16 MB of memory. It has a 1.8-inch Polysilicon TFT LCD display with 110,000 pixel resolution. The camera has USB and video out connectivity and is powered by four AA batteries.

It's a Match!



A digital camera records information about the settings used to capture an image. The digital image can then be transferred to your hard disk or a CD along with this information. When printing the image, the PIM-enabled printer reads the information to output a near-perfect match of the original image

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